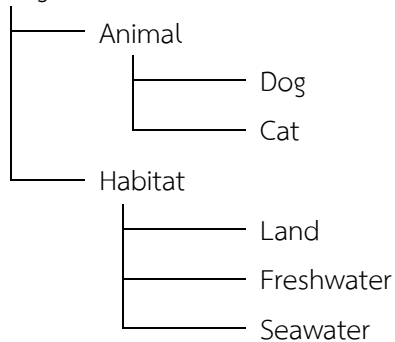


Instruction

Explanation of symbols used for representing the ontology

To describe ontology, please use the following format to represent class-subclass relationships.

- Things



To describe properties in the ontology:

Class (as Domain)	Property name	Range	Characteristics
Animal	hasName	string	-
	hasBirthdate	date	Functional
	hasHabitat	Habitat	-
Habitat	hasName	string	-
	isHabitatOf	Animal	-

To define class restrictions for Class A:

A
SubclassOf
 Pizza
Prop₁ only A
Prop₂ some B
Prop₃ value k
Prop₄ exactly 1 Integer
Prop₅ min 2 C
Prop₆ max 5 D
not (Prop₇ some B)

A
EquivalentTo
 Pizza
 and (*Prop₁ only A*)
 and (*Prop₂ some B*)
 and (*Prop₃ value k*)
 and (*Prop₄ exactly 1 Integer*)
 and (*Prop₅ min 2 C*)
 and (*Prop₆ max 5 D*)
 and (*not (Prop₇ some B)*)

Syntax for defining restriction on value

Example: **hasXX** some Integer [≥ 400]

- **Explanation of table and query**

To write SPARQL query, each table represents the data schema together with a few examples of the data. The label of each column represents property name. Data in “Object Label” column is the domain of all the properties represented in the rest of the column in each table.

For example:

Object Label	Instance_of	Country	City	Height
Shanghai Tower	skyscraper	China	Shanghai	632
Burj Khalifa	skyscraper	United Arab Emirates	Dubai	828
Merdeka 118	skyscraper	Malaysia	Kuala Lumpur	678.9

```
SELECT ?typeLabel ?countryLabel ?cityLabel ?height
```

```
WHERE
```

```
{   wd:Shanghai_Tower   wdt:Instance_of ?type;
    wdt:Country ?country.
    wdt:City ?city.
    wdt:Height ?height
```

```
SERVICE wikibase:label { bd:serviceParam wikibase:language "[AUTO_LANGUAGE]". }
```

This will return "skyscraper China Shanghai 632" as output.

- **Explanation of prefix used in the SPARQL for wiki data**

Suppose that the data in the table below are the instances of the Person-Job ontology

Person	Nationality	Gender	JobName
Nida_Somsuk	Thai	female	Professor JobPlace: Kasetsart_U ; Salary: 50,000
			Director JobPlace: OSC ; Salary: 35,000

wdt:<property name> for referring to property

For example: **?a wdt:Nationality ?b** // bind ?a to *Nida_Somsuk* ; ?b to *Thai*

?a wdt:Gender ?c // bind ?a to *Nida_Somsuk* ; ?c to *female*

wd:<class or instance name> for referring to class or instance

For example: **?a wd:Nationality wd:Thai** // bind ?a to *Nida_Somsuk*

p:<property name> for referring to a property where the instance that is the value of that property is a complex instance and has its own properties. **p** must be used when you also want to refer to such complex instance's properties.

To refer to the value of the complex instance that is the value of **p:<property name>**

ps:<property name> Use for referring to the actual value of the property called by **p**.

pq:<property name> Use for referring to other properties of the instance that is the value of property called by **p**

For example: **wd:Nida_Somsuk p:JobName ?jobstruct**

?jobstruct ps:Jobname ?a → ?a = Professor ; ?a = Director

?jobstruct pq:JobPlace ?b → ?b = Kasetsart_U ; ?b = OSC

?jobstruct pq:Salary ?c → ?c = 50,000 ; ?c = 35,000

- Query Example (for SPARQL syntax reference)

```
SELECT ?person ?personLabel ?dob
WHERE {
  ?person wdt:P31 wd:Q5;
          wdt:P569 ?dob.
  FILTER("2015-01-01"^^xsd:dateTime <= ?dob && ?dob < "2016-01-01"^^xsd:dateTime).
  SERVICE wikibase:label { bd:serviceParam wikibase:language "[AUTO_LANGUAGE]". }
}
```

```
SELECT ?country (MAX(?population) AS ?maxPopulation)
WHERE {
  ?city wdt:P31/wdt:P279* wd:Q515;
        wdt:P17 ?country;
        wdt:P1082 ?population.
}
GROUP BY ?country
```

```
SELECT ?material ?materialLabel (COUNT(?painting) AS ?count)
WHERE {
  ?painting wdt:P31/wdt:P279* wd:Q3305213;
            p:P186 ?matstruct.
  ?matstruct ps:P186 ?material;
            pq:P518 wd:Q861259.
  SERVICE wikibase:label { bd:serviceParam wikibase:language "[AUTO_LANGUAGE]". }
}
GROUP BY ?material ?materialLabel
ORDER BY DESC(?count)
LIMIT 20
```

```
SELECT ?country ?countryLabel (COUNT(?lang) AS ?numberOfLanguage)
WHERE {
  ?country wdt:P31 wd:Q6256;
           wdt:P2936 ?lang.
  SERVICE wikibase:label { bd:serviceParam wikibase:language "[AUTO_LANGUAGE], mul, en" }
}
GROUP BY ?country ?countryLabel
ORDER BY DESC(?numberOfLanguage)
```

```
SELECT ?person ?personLabel ?start ?end ?birth ?death ?death_age
WHERE {
  ?person wdt:P31 wd:Q5;
          p:P39 [ps:P39 wd:Q11696; pq:P580 ?start; pq:P582 ?end ].
  OPTIONAL {?person wdt:P569 ?birth}
  OPTIONAL {?person wdt:P570 ?death}
  BIND(YEAR (?death) - YEAR (?birth) AS ?death_age )

  SERVICE wikibase:label { bd:serviceParam wikibase:language "[AUTO_LANGUAGE], mul, en" }
}
LIMIT 100
```

- **List of aggregate function:**

COUNT : the number of elements. You can also write COUNT(*) to simply count all results.

SUM, AVG: the sum or average of all elements, respectively. If the elements aren't numbers, you'll get weird results.

MIN, MAX : the minimum or maximum value of all elements, respectively. This works for all value types; numbers are sorted numerically, strings and other types lexically.

SAMPLE : any element. This is occasionally useful if you know there's only one result, or if you don't care which one is returned.